National College of Ireland

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Cloud Machine Learning

MSCCLOUD1_A and MSCCLOUD1_B

Release Date: 28 / 01 / 2026

Submission Deadline:
Wednesday, 13th April 2026 at 23:55 hrs.

Continuous Assessment (CA) Type: Project 50%

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Weight: The assignment will be marked out of 100.

Instructions: Work in groups of 3-4 students

SUBMISSION DETAILS:

Report:

1. A report of up to 5 pages that addresses the assignment is required to be submitted online on Moodle before the deadline. There will be a 10% penalty for each extra page.
2. The report must be formatted according to the IEEE - Manuscript Templates for Conference Proceedings, A4 page size, and must be submitted as a PDF file.
3. Make sure to acknowledge the original sources for your investigation as appropriate including the use of AI. You MUST include the NCI assignment submission cover page providing this information if applicable.
4. The assignment submission coversheet does not count in the 5-page limit.
5. Everything else, such as references, appendices etc. is included in the 5-page limit.
6. Only ONE team member, whose surname is alphabetically the first, MUST upload the report.

Source code and dataset(s) used in the project:

The source code and dataset(s) used in the project must be submitted online on Moodle before the deadline as a single compressed .ZIP file.

Presentation:

1. **4-5 minutes** that include a demonstration of your work.
2. **The presentation/demo will be held in class** after the submission deadline.
3. If you exceed the time limit of 5 minutes, a 20% deduction per additional 30 seconds will be applied.
4. **All team members MUST be present**. Those who are absent during the presentation will be marked absent and will be awarded ZERO marks.

TURNITIN: All report submissions will be electronically screened for evidence of academic misconduct (i.e., plagiarism and collusion)

Project Brief

1. Introduction

This project is aimed at achieving the module Learning Outcomes listed below:

LO2	Clean and transform datasets in preparation for data mining, and build and evaluate machine learning models to extract knowledge from various datasets.
LO4	Evaluate and utilise cloud computing technologies and services for data collection, storage and mining when designing and implementing data driven applications.

2. Project Description

Students are required to propose and execute a research project using **machine learning techniques in the scope of the course as a team**. The project should relate to a real-world problem in any topic. Students should make use of cloud computing technologies and services for **data collection, storage and mining**. While doing the project, students should consider the **ethical aspects** with regard to the datasets and machine learning algorithms used.

The overarching focus of the project is to **develop a cloud-based solution of a problem with machine learning methods/techniques**. Your solution should aim at roughly balancing the three paradigms, namely performance, scalability and latency and throughput, of your models. The development pipeline should include the following aspects: data collection process in cloud, transformation strategies applied (locally or in cloud). In order to view data characteristics, you should apply statistical analysis tools. The expectation is that you will be applying a number of ML techniques and models in your experiments, and each method should be applied to answer **specific research question(s) aligned to the overall goal(s)** of the project. It is also expected that the application of each method is accompanied by an appropriately sized review **documenting** pertinent and contemporary approaches in the literature that can both inform the application of a method as well as **justify its potential merit(s)**. You must use **appropriate evaluation techniques and tools** in order to measure the performance (in terms of quality and latency) of your methods / models. You should **perform** a comprehensive manual error analysis on the outputs produced by your models and baseline models which have been considered for your investigation.

Projects will be assessed based on their **novelty, technical quality, insightfulness, depth, clarity, robustness, quality of writing and reproducibility**. Code and datasets (different for each team) must also be submitted with the report. Algorithms and resources used in a report should be described as completely as possible to allow reproducibility. This includes experimental methodology, empirical evaluations, and results. The reproducibility factor will play an important role in the assessment of each submission.

It is essential that the project shows unambiguously evidence of:

1. A critical analysis of fundamental machine learning methodologies to assess best practice guidance when applied to computational problems (e.g. data mining, regression, classification, etc.) in the specific context of the project.

2. The extraction, transformation, exploration, and cleaning of datasets in preparation for building machine learning methods used in the project.
3. Making use of cloud computing technologies and services for data collection, storage and mining.
4. The building and evaluation of machine learning models on a variety of datasets and parameters; producing learning curve of your ML models on varying sized datasets, data domains and number of parameters.
5. Reporting performance (quality, latency and throughput) and scalability of the machine learning models in varying-sized data sets. The discussion of performance should be orientated around multiple notions of performance and across a variety of parameters (e.g. latency).
6. The extraction, interpretation, and evaluation of information and knowledge that is drawn from the datasets used.
7. The critical review of relevant machine learning research to afford the assessment of research methods applied in the project.
8. A prototype showing facilitating services (e.g. real-time or batch) with your models via cloud to the end-users.

3. Report Guidelines

The final report must follow the IEEE conference format and should be between 4–5 double column pages in length (this includes all figures, but not references).

For this report IEEE referencing style should be used. Microsoft Word and LATEX templates are available at http://www.ieee.org/conferences_events/conferences/publishing/templates.html

The following structure is suggested for the report:

Abstract: 100–150 words providing a high-level description of the project, its core findings, and the domain of the datasets (not necessarily in this order).

Introduction: Remainder of 1st page. It should motivate the work, present and discuss the research question(s) / objective(s) of the project and (optionally) provide a concise overview of the following sections (max 1–2 lines per each).

Related Work: Half page -– this should not only summarize the related works, but also critically evaluate their key positive and negative aspects with respect to the topic and domain of the project, i.e., how well/badly do the related works artefact address your question(s) / objective(s), what aspects are useful to consider, what are the limitations, etc. Also include here a discussion on the previous uses of the datasets and the methods applied. If you plan to reuse a method already applied to this dataset, discuss what you expect to gain by doing this.

Methodology: This section can be named differently. But it should describe how have you approached answering your question. Additional (technical) details can also be discussed here. Essentially, you should recount how you applied ML models in the cloud to facilitate your research question(s). You should also include here a discussion on key preliminary aspects of the methodology, such as how the datasets have been prepared for study (i.e., the pre-processing, and transformation stages).

Evaluation: How have you used your methodology to answer the question (evaluation methodology), i.e., how do you know that a method is good? what performance measures you have selected and why (discuss how the choice of performance measures is appropriate). If you have to parametrize part of an approach, how have you done that, and why were these choices made, and what impacts

can different parameters have on your results? You should also discuss the results in detail in this section: what are their implications? What do they show / not show? etc. A discussion on sampling methods is expected here too. This section should include a subsection 'error analysis' which will present a thorough and comprehensive error analysis on the outputs of your systems / models including the baselines.

Conclusions and Future Work: Summarize your findings and discuss limitations / extensions you would do next to improve or extend your study if you had more time. Summarize the (partial) answer to the research question(s) at a high level, and note the key implications of your findings with respect to the methods studied.

Contributions: A table should be used in order to show individual contributions in the project. The table should clearly highlight who carried out what part of the project (e.g. design, development, experiments and report preparation).

References: Include a list of all references used in your report.

Although not recommended, you may remove, change and/or alter methods, datasets and/or key research question(s) or objective(s) after submission of the Proposal / Interim Progress Report.

4. Marking Grid

This project will be graded using the marking grid shown in Table 1. Note that marks in Table 1 represent percentage. THE FINAL MARK MUST BE 40% OR ABOVE TO ACHIEVE A PASS.

Extension/Re-run

Should any student miss the assessment deadline with a valid reason, he/she can now apply for an application for coursework Extension/Re-run Form online, via NCI360.

N.B.

All submissions will be electronically screened for evidence of academic misconduct (plagiarism and collusion).

Table 1. Grading Rubric

	0 – 39	40 – 49%	50 – 59%	60 –69%	70% +	Marks
PROJECT SPECIFICATION 10% **Project Objectives: definition & attainment**	Objectives are inadequately specified	Objectives may be specified but not achieved.	Objectives have been clearly specified and achieved to some extent.	Objectives have been clearly specified and are appropriate. The objectives have been fully achieved.	Objectives have been clearly specified and are creative and appropriate. Objectives have been fully achieved or surpassed	0-10
LITERATURE REVIEW 10% **Use of and critical understanding of relevant theory**	Very limited use of theory and concepts	Limited use of theory and concepts	Reasonable use of theory and concepts but lack of breadth in literature reviewed.	Comprehensive use of relevant theory and concepts. Evidence of breadth in literature review.	Critical application and critique of relevant theory and concepts. Evidence of breadth and depth of literature reviewed.	0-10
ARTEFACT / PRODUCT DEVELOPMENT 30% **Relevance and justification of methodology utilised and the ability to carry out the specified methodology**	Alternative methodologies not considered and method selected not justified. Significant errors in the application of methodology.	Alternative methodologies have not been considered and only limited justification for selected methods. Some errors in application.	Alternative methodologies have not been fully considered, though a reasonable attempt made at justifying its use. The methodology has been completely carried out.	Alternative methodologies have been considered and the methodology selected has been justified. The methodology has been competently carried out.	Alternative methodologies have been fully considered and the chosen methodology fully justified. The application has been rigorously carried out.	0-30
EVALUATION 25% **Analysis of the research findings and ability to understand the study's implications and limitations**	Pedestrian or poor attempt to analyse the findings. Implications and limitations not understood.	Some attempt at analysis although with some problems. Implications and limitations not well understood.	A reasonable attempt to analyse the findings. However, the synthesis of data collected and theory is not fully achieved. Implications not fully appreciated.	Rigorous analysis of findings. Demonstrates the ability to synthesise data collected and relevant theory. Shows an understanding of implications of the study.	Rigorous and creative analysis of findings. Demonstrated the ability to synthesise data collected and relevant theory. Insightful conclusions which appreciate limitations and implications of the study.	0-25
DOCUMENT STRUCTURE, PRESENTATION & REFERENCING 15% **Quality of** document **prese**ntation – including referencing, **structure**, grammar, and spelling.	Poor presentation with inconsistent referencing. Inadequate grammar and spelling.	Reasonable presentation but inconsistent in referencing. Acceptable grammar and spelling.	Satisfactory presentation with consistent referencing and clear structure. Acceptable grammar and spelling.	Good presentation and structure with rigorous referencing. Correct grammar and spelling	Excellent presentation and structure with rigorous referencing. Correct grammar and spelling.	0-15
VIVA 10% **Quality of** Oral **Prese**ntation – including handling of questions from examiners	Unacceptable oral presentation and handling of questions.	Poor oral presentation and weak handling of questions.	Neat oral presentation and acceptable handling of questions.	Clearly presentation with good handling of questions.	Excellent well directed presentation with impeccable handling of questions.	0-10